

Krishika Garg

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EDUCATION

Manipal Institute of Technology

B.Tech. in Computer Science and Engineering(Cyber Security)—CGPA: 8.27

Manipal, India

2022 – 2026

D.N. Model

Senior Secondary (Class XII)—86%

Moga, India

2020 – 2022

TECHNICAL SKILLS

Languages: Java, C/C++, Python, SQL, HTML, CSS

Frameworks & Technologies : Spring Boot, REST APIs, Data Structures, Power BI

Security Tools : Kali Linux, Splunk, Wireshark

EXPERIENCE

Intern | IIT Madras Pravartak

March 2025 – June 2025

- Engineered secure **data pipeline** for crop disease detection using multispectral imagery, improving classification accuracy.
- Performed vulnerability assessments; implemented **AES encryption** and **SSL/TLS protocols** across distributed networks, reducing attack surface by 35%.

PROJECTS

Shipping Cost Calculation Microservice | Java, Spring Boot, H2, Maven, Postman

March 2026

- Designed and developed a **Spring Boot**-based **microservice** with **5+ REST APIs**, implementing layered architecture (Controller, Service, Repository) and **Spring Data JPA**, reducing code coupling by ~40%.
- Implemented **Haversine-based distance pricing** and optimized nearest warehouse selection across **4+ entities**, improving routing efficiency and reducing computation time by ~30%.
- Built and tested APIs using **Postman (10+ test cases)**; implemented **@RestControllerAdvice** for handling **8+ edge cases**, improving system reliability by ~25%.

Web Penetration Testing Tool (CLI-Based) | Kali Linux

July 2025

- Built a **command-line web vulnerability scanner** with **6+ automated modules** for **SQL Injection**, **XSS**, **SSL/TLS misconfigurations**, **authentication bypass**, and **API fuzzing**
- Enhanced **scan efficiency by 40%** through **asynchronous TCP/UDP port scanning** and integrated **auto vulnerability classification** with detailed **HTML security reports**.
- Discovered and validated **10+ critical web application vulnerabilities** across test environments using **custom payloads**, **rockyou.txt wordlist**.

D2C Product Management Web Application | React.js, Node.js, MongoDB

August 2025

- Built a responsive web application using **React.js** to manage product listings for a D2C fashion brand, including adding, viewing, updating, and deleting products.
- Implemented backend logic using **Node.js** with direct database interaction to store and retrieve product data from **MongoDB**.
- Integrated frontend and backend using **REST-style API calls** and followed Git-based version control for code management and collaboration.

Federated Learning Under Data Poisoning Attacks | Python, PyTorch, FLwr, Pandas, NumPy

April 2025

- Analyzed **70K-record** federated dataset across 10 clients; built preprocessing, partitioning, augmentation pipelines for validation and anomaly detection.
- Simulated poisoning attacks over **50+ rounds**; benchmarked **FedAvg**, **Krum**, **FoolsGold**, reducing ASR from **37% to 10%** while maintaining **95% accuracy**.

POSITION OF RESPONSIBILITY

Core Committee Member | Drama Club

August 2023 – July 2024

- Led **200+ members**, **organized 10+ events**, and managed operations with a **50+ member team**, ensuring timely execution and representing the club in **5 intra-institute activities**.